

Nisip Sigdel Computer Science Graduate | AI, NLP & Machine Learning

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SUMMARY

Computer Science graduate with hands-on experience in Natural Language Processing, Machine Learning, Large Language Models, and Retrieval-Augmented Generation. Developed and contributed to AI systems involving transformer-based NLP models, semantic search, vector databases, and locally deployed LLMs. Interested in research focused on NLP, LLMs, information retrieval, and intelligent systems for understanding and processing real-world language and document data.

EDUCATION

B.Sc. Computer Science & Information Technology

Tribhuvan University

04/2022 – 04/2026

Kathmandu, Nepal

Completed all 8 semesters of the BSc CSIT program in 2026. Contributed to an undergraduate group project focused on NLP-based resume screening and job recommendation, with primary involvement in dataset preparation and transformer-based model training using spaCy, RoBERTa, and Google Colab.

RESEARCH INTERESTS

AI, Machine Learning & NLP

LLMs | RAG | Information Retrieval | Intelligent Systems

Natural Language Processing · Large Language Models · Retrieval-Augmented Generation · Information Retrieval · Semantic Search · Machine Learning · Intelligent Systems · AI for Document Understanding

PROJECTS

AI PDF Reader & Autonomous Finance RAG Assistant 🔗

07/2026 – 08/2026

RAG · LLMs · NLP · Information Retrieval · Python · Ollama · ChromaDB

- Developed a Retrieval-Augmented Generation (RAG) system for question answering over user-provided PDF documents using document parsing, text chunking, embeddings, vector retrieval, and context-aware generation.
- Implemented semantic retrieval using Hugging Face embeddings with **ChromaDB** and **FAISS**, enabling relevant document context to be retrieved before LLM generation.
- Integrated **Llama 3.1 8B** through **Ollama** for local LLM inference, enabling document analysis without relying on external LLM APIs.
- Developed an automated financial information pipeline using **Google News RSS**, **BeautifulSoup**, **yfinance**, and **scheduled background tasks** to collect and index current financial information.
- Designed retrieval and prompting constraints to keep generated responses grounded in retrieved context and avoid direct investment recommendations.
- Built a modular **Streamlit** interface with PDF analysis, finance dashboard, and chat functionality, supported by **SQLite logging** and **Pytest-based testing**.

NLP-Based Job Recommendation & Resume Screening System 🔗

NLP Model Training | spaCy Transformers | RoBERTa | Google Colab

- Contributed to an undergraduate group project focused on NLP-based resume screening and job recommendation.
- Prepared and processed training data and configured the NLP pipeline for model training.
- Trained a transformer-based Named Entity Recognition (NER) model using spaCy Transformers and RoBERTa-base in Google Colab.
- Tested and evaluated the trained model and successfully ran the saved model for resume entity extraction.
- Saved the trained model for potential integration into the broader application.

SIP N DIP — Business Website 🔗

Client Project | React.js | Node.js | Express.js | MongoDB

- Collaborated with a development team at **Mansa Tech Group** to build a production website for SIP N DIP, a New Zealand-based food and beverage business.
- Contributed to the development of responsive and user-focused interfaces using **React.js**.

- Developed and integrated backend functionality using **Node.js** and **Express.js**, with **MongoDB** for data management.
- Worked with team members to implement client requirements, troubleshoot issues, test functionality, and improve the overall user experience.

GameVault

Full-Stack Web Application | React | Node.js | Express | PostgreSQL

- Developed a full-stack web application for sharing and discovering links to **legally distributable community-created game resources**, including patches, save files, custom maps, and mods.
- Built the frontend using **React and Vite**, with a modular component architecture and **mobile-first responsive design**.
- Developed backend services using **Node.js and Express**, with **PostgreSQL hosted on Railway** for database management and API integration.
- Implemented **user authentication, search and filtering, comments, ratings, and an administrative dashboard**, with Cloudinary-based image handling and Vercel deployment.

PROFESSIONAL EXPERIENCE

Web Developer & AI Content Contributor

Digital Idea2Grow 

07/2024 – Present
Biratnagar, Nepal

- Developed and maintained the **Idea2Grow website**, contributing to its structure, responsive design, functionality, and overall user experience.
- Contributed to writing and developing technical content related to **Artificial Intelligence, emerging AI technologies, and digital marketing**.
- Helped the team **optimize client and company content for SEO**, improving search visibility and content structure.
- Researched and applied **SEO and Answer Engine Optimization (AEO)** practices to articles and web content.
- Collaborated with the team on **website updates, content development, keyword research, and online visibility improvements**.

Software Developer (Freelance)

Mansa Tech Group 

04/2025 – Present
Remote

- Developed responsive web applications and user-focused interfaces for business clients.
- Built and integrated frontend components with backend services through **REST APIs** for dynamic data exchange and application functionality.
- Worked across frontend and backend development to deliver complete web-based solutions tailored to client requirements.
- Collaborated on application development and deployment while troubleshooting issues and improving system functionality and reliability.

Quality Assurance Intern

Elite Technologies 

10/2025 – 12/2025
Biratnagar, Nepal · Remote

- Analyzed application requirements and designed test cases covering **UI, navigation, form validation, and core functionality**.
- Executed **manual, regression, and cross-browser testing** to identify functional and usability issues.
- Documented and reported software defects using **Jira**, including reproduction steps, expected and actual results, and severity.
- Maintained test case documentation and testing records using **Microsoft Excel**.
- Collaborated with development and design teams to communicate issues and support software quality improvements.

SKILLS

AI / Machine Learning

Natural Language Processing (NLP) · Large Language Models (LLMs) · Retrieval-Augmented Generation (RAG) · Information Retrieval · Semantic Search · Transformer Models · Machine Learning · spaCy · RoBERTa · Scikit-learn

Software Development

React.js · Node.js · Express.js · PostgreSQL · REST APIs · HTML · CSS · JavaScript · C++ · C# · C

AI / Data Tools

Python · Pandas · NumPy · Matplotlib · Jupyter Notebook · Hugging Face · LangChain · Ollama · ChromaDB · FAISS · Streamlit · SQLite · Pytest

Testing & Other

Manual Testing · Functional Testing · Regression Testing · Jira · WordPress · SEO · AEO

CERTIFICATES

Python Data Structures

University of Michigan · Coursera · March 2023

Programming for Everybody

University of Michigan · Coursera · January 2023

AWARDS AND LEADERSHIP

Intra-College Sports Event — Birat Multiple College

Birat Multiple College

Organized the college's Chess Tournament — managed scheduling and coordinated participating teams — as part of an intra-college sports competition open to students across all semesters.

Also competed across multiple sports at the same event: 1st place in Carrom Board, 2nd place in Chess, Cricket, Volleyball, and Football.

LANGUAGES

English

Nepali

Hindi